

# Lecture 4: Consumer Choice (Chapter 5)

*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, Chapter 5 — Consumer choice (§5.1 modelling choice · §5.2 preferences & utility · §5.3 budget constraints · §5.4 utility maximization · §5.5 discussion of the framework · §5.6 behavioral economics · §5.7 appendix: solving for the optimal basket with Lagrange). *Lecturer: Luigi Butera — deck "Lecture 4: Consumer choice."* (Numbering note: this is the course's 4th lecture but the textbook's Chapter 5; the deck's own "4.1–4.5" roadmap just re-numbers the chapter's sections.)

**Theme of the lecture (Butera — from the deck).** So far (Lectures 2–3) we took **demand curves as given** and pushed them around. Now we **build demand from the ground up**: *where do preferences come from, and how does a rational consumer choose under scarcity?* The deck's answer is a **three-stage machine** — **preferences → utility → budget constraint → constrained optimization** — that will (next chapter) *generate* the demand curve. The deck's five stops: **4.1 preferences · 4.2 utility · 4.3 budget constraints · 4.4 constrained choice · 4.5 behavioral economics.**

★ **The heart of the lecture:** a consumer **maximizes utility subject to a budget constraint**. We (1) describe what she **likes** with well-behaved **preferences** (complete + transitive), (2) quantify them with a **utility function** whose slope-of-indifference-curve is the **MRS** = how much of one good she'll trade for the other *and stay equally happy*, (3) describe what she **can afford** with a **budget line** whose slope is the **MRT** =  $p_F/p_E$  = how the market lets her trade the goods. The **optimum** is where the highest reachable indifference curve is **tangent** to the budget line — i.e. where **MRS = MRT** ( $MU_F/MU_E = p_F/p_E$ ): the rate she's *willing* to trade equals the rate the market *lets* her trade.

**The red thread.** Every piece answers one of three questions: *what do I want?* (preferences → utility → MRS), *what can I afford?* (budget line → MRT), and *so what do I pick?* (tangency: MRS = MRT). Get those three and you can predict how choices shift when **prices** or **income** change — which is exactly a **demand curve** (next chapter). §5.6 then asks: *when does this rational machine break down?* (behavioral economics).

## Key concepts / "modes" to use

- **Bundle** — a list of quantities of each good (e.g. "4 Entertainment, 2 Food"). Choice = ranking bundles.
- **Preferences & the rationality axioms** — **completeness** (can rank any two bundles) · **transitivity** (rankings "hold in transit") · **monotonicity** / "more is better" (convenient add-on).
- **Utility function**  $U(F, E)$  — assigns a number to each bundle so that **higher = preferred**. **Ordinal**, not cardinal (only the *ranking* means anything).
- **Marginal utility (MU)** — extra utility from one more unit of a good, holding the other fixed = the **partial derivative**; typically **diminishing**.
- **Indifference curve** — all bundles giving the *same* utility; **5 properties**; three shapes (**perfect substitutes, imperfect substitutes, perfect complements**).
- **Marginal rate of substitution (MRS)** — slope of the **indifference curve** =  $MU_F / MU_E$  = how much E you'll give up for one more F at equal utility.

- **Budget constraint / line** —  $I = p_F \cdot F + p_E \cdot E$ ; the **opportunity set** is everything on or under it.
- **Marginal rate of transformation (MRT)** — slope of the budget line =  $p_F / p_E$  = how the *market* lets you swap the goods.
- **Consumer optimum** — highest indifference curve tangent to the budget line: **MRS = MRT**, i.e.  $MU_F / MU_E = p_F / p_E$  (equivalently, equal marginal utility *per euro*:  $MU_F / p_F = MU_E / p_E$ ).
- **Interior vs corner solution** — tangency inside the axes vs "spend everything on one good" (perfect substitutes).
- **Behavioral economics** — systematic deviations: **social preferences, lack of self-control, cognitive biases** (loss aversion, endowment effect, framing, overconfidence).

## Section-by-section main points

### §5.1 How can we model choice? — the three ingredients (deck)

**Core:** we model an individual who **faces constraints** (income, time) and **does as well as possible** given them. Sounds almost tautological — the whole chapter is about *putting structure* on "does as well as possible." The deck lists the **ingredients we need**:

- A language to say "I like this more than that" → **preferences** (§5.2).
- **Assumptions** that make those preferences *rational* → the **axioms** (completeness, transitivity).
- A tool to quantify wellbeing → the **utility function** (§5.2.1).
- A tool to quantify "you can't have it all" → the **budget constraint** (§5.3).
- Bringing it together → **constrained optimization** (§5.4).
- Three premises of the whole build (deck + §5.1): (1) we know and have stable preferences, (2) we face constraints (time, money), (3) we try to do as well as possible given them.

### §5.2 Preferences and the rationality axioms (deck slides 6–7)

**Core:** choice is modelled as **ranking bundles**. For the ranking to give a consistent model we assume it obeys two **essential** axioms (plus one convenient one). The deck's notation for the ranking:

| Symbol     | Reads                                 | Meaning                           |
|------------|---------------------------------------|-----------------------------------|
| $a > b$    | "a is <b>strictly preferred</b> to b" | a is <i>better</i> than b         |
| $a \geq b$ | "a is <b>weakly preferred</b> to b"   | a is <i>at least as good</i> as b |
| $a \sim b$ | " <b>indifferent</b> between a and b" | a and b are <i>equally good</i>   |

- **Completeness (essential):** the consumer can **rank any two bundles** — for any A, B she can say  $A > B$ ,  $B > A$ , or  $A \sim B$ . (No "I can't compare these.")
- **Transitivity (essential):** rankings "**hold in transit**" — if  $A > B$  and  $B > C$ , then  $A > C$ . This is what lets us predict choice in a *chain* of comparisons (and stops preferences from cycling).
- **Monotonicity / "more is better" (convenient, not essential):** all else equal, **more of a good is better**. This is what makes bundles *farther from the origin* preferred. (Things you want *less of* — pollution, noise — are relabelled "*bads*," and we redefine the good as their *absence*.)

- **Why the axioms matter:** completeness + transitivity are *exactly* the conditions under which preferences can be represented by a utility function (next section). No axioms → no utility function → no model.

## §5.2.1 Utility & utility functions (deck slides 8–9)

**Core:** if preferences are **complete and transitive**, we can represent them with a utility function  $U(\cdot)$  that assigns each bundle a **number**, higher for more-preferred bundles. Neoclassical economics inherits this from the **utilitarian** tradition (J.S. Mill: "*actions are right in proportion as they tend to promote happiness*").

### UTILITY REPRESENTS PREFERENCES

$$a \succ b \iff U(a) > U(b)$$

*Reads:* the utility function just **translates the ranking into real numbers** — bundle  $a$  is preferred to  $b$  if **and only if**  $a$  scores a higher utility number. (This is the **von Neumann–Morgenstern** representation the deck cites.) The function *orders* preferences; it does not measure "happiness" in any absolute unit.

- **Ordinal, not cardinal (the crucial caveat):** only the **ranking** matters, **not** the size of the gap.  $U = 8$  vs  $U = 4$  means the first is **preferred** — it does **not** mean "twice as good." (Deck: " $9 > 1$ , but do '1' and '9' mean something meaningful?" — no. Utility is unobservable.)
- **General vs specific form:** general  $U = U(F, E)$ ; a specific example the textbook uses is  $U = \sqrt{EF}$ . **Worked:** at  $E=2, F=8$ ,  $U = \sqrt{(2 \cdot 8)} = \sqrt{16} = 4$ . Bundles with the **same product**  $E \cdot F = 16$  (e.g.  $E=4, F=4$  or  $E=8, F=2$ ) sit on the **same** indifference curve;  $E=5, F=5$  gives  $U = 5 \rightarrow$  a **higher** curve.
- **Common functional forms (deck slide 12):**
  - **Cobb-Douglas:**  $U = F^a \cdot E^b$  — the workhorse; convex, imperfect substitutes.
  - **Perfect substitutes:**  $U = a \cdot F + b \cdot E$  — straight-line indifference curves.
  - (Perfect complements:  $U = \min(F, E)$  — L-shaped.)

## ★ Marginal utility & diminishing marginal utility (deck slides 10, 16)

**Core:** marginal utility (MU) is the extra utility from **one more (infinitesimal) unit** of a good, holding the others fixed — i.e. the **partial derivative** of  $U$ . It is usually **diminishing** (the 1st slice of pizza beats the 7th).

### MARGINAL UTILITY

$$MU_F = \partial U(F, E) / \partial F \quad \cdot \quad MU_E = \partial U(F, E) / \partial E$$

*Reads:* **how much total utility rises per extra unit of one good, other goods held constant** —  $\partial U / \partial F$  = the partial derivative w.r.t. Food • **diminishing MU** = this derivative *shrinks* as you get more of the good.

- **The deck's three consumers (utility over one good):** for each extra apple —
  - **Bob:** each apple adds the **same** utility → **constant** MU (linear  $U$ ).
  - **Linda:** each apple adds a bit **more** → **increasing** MU (convex  $U$ ). (*unusual*)

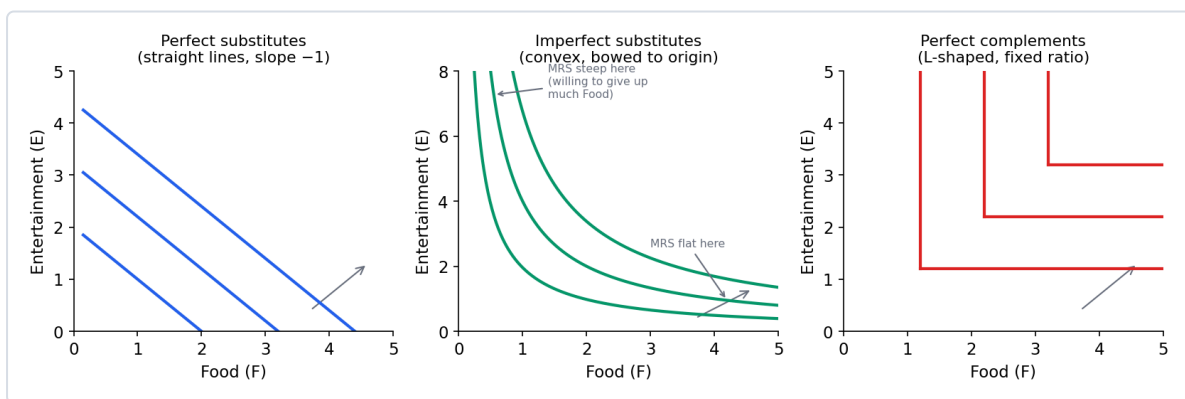
- **Mike:** each apple adds a bit **less** → **diminishing MU** (concave U). (*the normal case*)
- **Worked example** ( $U = \sqrt{EF} = E^{0.5}F^{0.5}$ ):  $MU_F = \partial U / \partial F = 0.5 \cdot E^{0.5} \cdot F^{-0.5} = 0.5 \cdot \sqrt{E/F}$ . At  $E=2, F=8$ :  $MU_F = 0.5 \cdot \sqrt{2/8} = 0.5 \cdot 0.5 = 0.25$ . As  $F$  rises (holding  $E$ ),  $\sqrt{E/F}$  falls →  $MU_F$  falls → diminishing marginal utility. ✓
- **Why it matters:** diminishing MU is *why* indifference curves are **bowed toward the origin** (convex) and *why* the **MRS falls** as you move down a curve — the next two ideas.

## Indifference curves — 5 properties & the 3 types (deck slides 13–14)

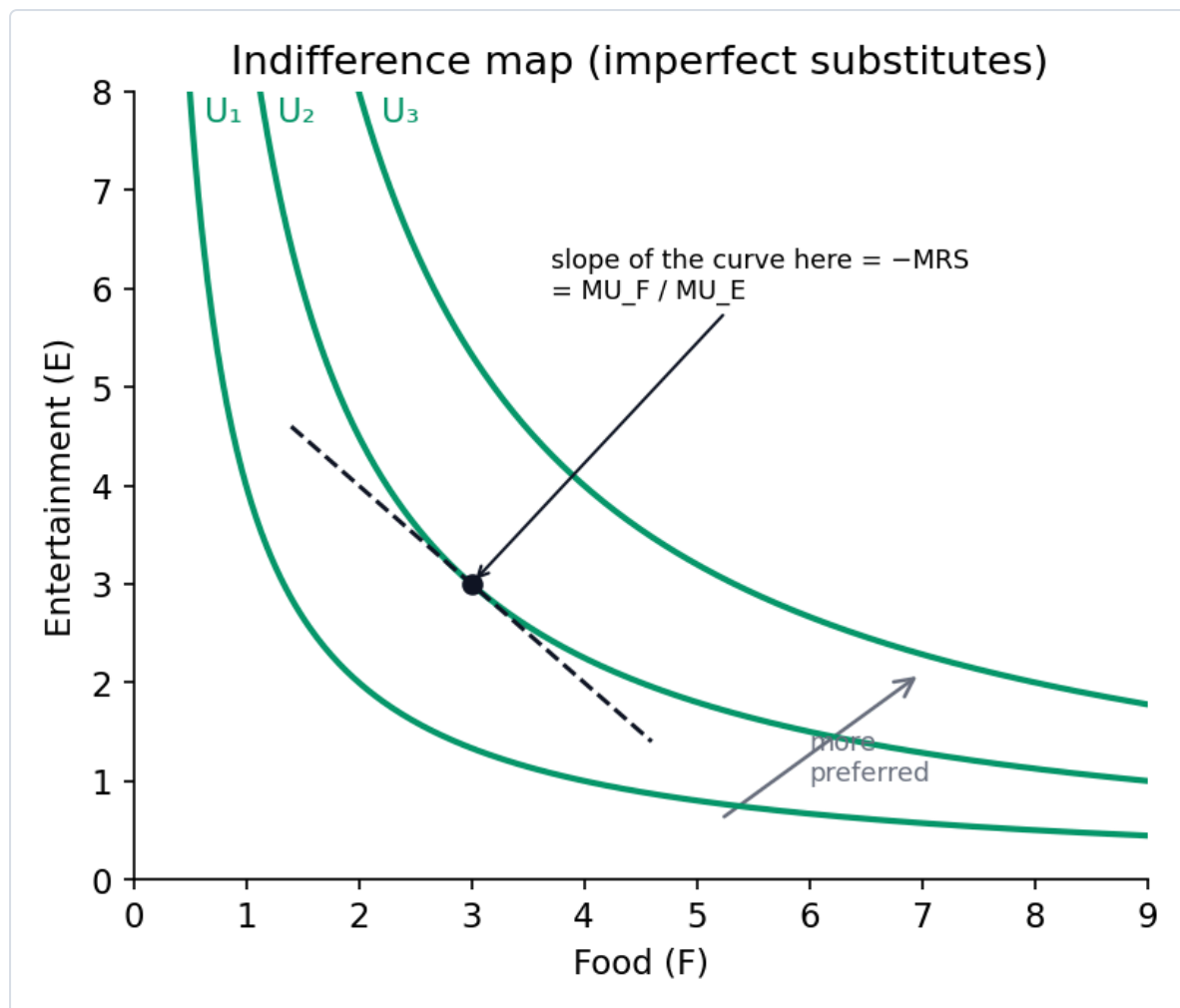
**Core:** an indifference curve connects all bundles the consumer views as equally desirable. It's the level-set of the utility function (fix  $U$  at one number). The deck's **five properties**:

| # | Property                                                   | Why                                                                |
|---|------------------------------------------------------------|--------------------------------------------------------------------|
| 1 | Curves <b>farther from the origin</b> are preferred        | "more is better" (monotonicity)                                    |
| 2 | <b>Every</b> bundle lies on some <u>indifference curve</u> | completeness                                                       |
| 3 | Indifference curves <b>cannot cross</b>                    | transitivity (crossing → contradiction)                            |
| 4 | They <b>slope downward</b>                                 | to keep utility fixed, more of one good ⇒ less of the other        |
| 5 | They <b>cannot be "thick"</b>                              | a thick band would contain a strictly-better bundle (monotonicity) |

- (Utility is read **ordinally**: the labels on the curves rank them; the gaps between labels are meaningless.)
- **Three shapes of preference** (textbook §5.2):



- **Perfect substitutes** — tradeable at a **constant rate** → **straight-line** curves (constant slope). *Example:* one €20 note ~four €5 notes, always.  $U = aF + bE$ .
- **Imperfect substitutes (the normal case)** — willing to trade, but the **rate depends on how much of each you already have** → **convex** curves, "bowed toward the origin" (strictly convex preferences). *When you have lots of chips and no drink, you'll give up many chips for one drink; when you have few chips, you won't.*  $U = \sqrt{EF}$ , Cobb-Douglas.
- **Perfect complements** — consumed in a **fixed ratio** → **L-shaped** curves. *Left & right shoes: a 2nd right shoe alone adds nothing.*  $U = \min(F, E)$ .



**How to read it (indifference map).** Each green curve is one utility level; **northeast = more preferred** (property 1). The dashed tangent at the marked bundle has slope  $-MRS$ : it gets **flatter** as you move right (more Food, less willing to give up Food) — that's diminishing MRS, the topic below.

## ★ Marginal rate of substitution (MRS) (deck slides 17–19)

**Core:** the **MRS** is the slope of the indifference curve — the maximum amount of one good the consumer will give up to get one more unit of the other, staying equally happy. Because utility is constant along the curve ( $dU = 0$ ), it equals the **ratio of marginal utilities**.

### MARGINAL RATE OF SUBSTITUTION

$$MRS = -dE/dF = MU_F / MU_E \quad (\text{good on the x-axis over good on the y-axis})$$

**Reads:** how much E you'll trade for one more F and stay on the same curve — derive it by **totally differentiating**  $U$  and setting  $dU = 0$ :  $0 = MU_F \cdot dF + MU_E \cdot dE \rightarrow -dE/dF = MU_F / MU_E$ .  $MU_F$  = marginal utility of the x-axis good (Food) ·  $MU_E$  = marginal utility of the y-axis good (Entertainment) · the ratio = the curve's steepness at that point.

- **Diminishing MRS (why the curve bows):** as you move **down-right** along the curve (more F, less E),  $MU_F$  falls and  $MU_E$  rises, so the ratio  $MU_F / MU_E$  **shrinks** — the curve **flattens**. "When I have relatively less pizza I'll give up more burritos for a slice than when I already have lots of pizza."

- **Worked example ( $U = \sqrt{EF}$ ):**  $MU_F = 0.5\sqrt{E/F}$ ,  $MU_E = 0.5\sqrt{F/E}$ , so  $MRS = MU_F/MU_E = (0.5\sqrt{E/F})/(0.5\sqrt{F/E}) = E/F$ . At bundle  $E=2, F=8$ :  $MRS = 2/8 = 0.25 \rightarrow$  she'll give up only **0.25 units of E** for one more F (she already has lots of F). At  $E=8, F=2$ :  $MRS = 8/2 = 4 \rightarrow$  now she'll give up **4 units of E** for one more F. *Same curve, opposite ends  $\rightarrow$  diminishing MRS.* ✓

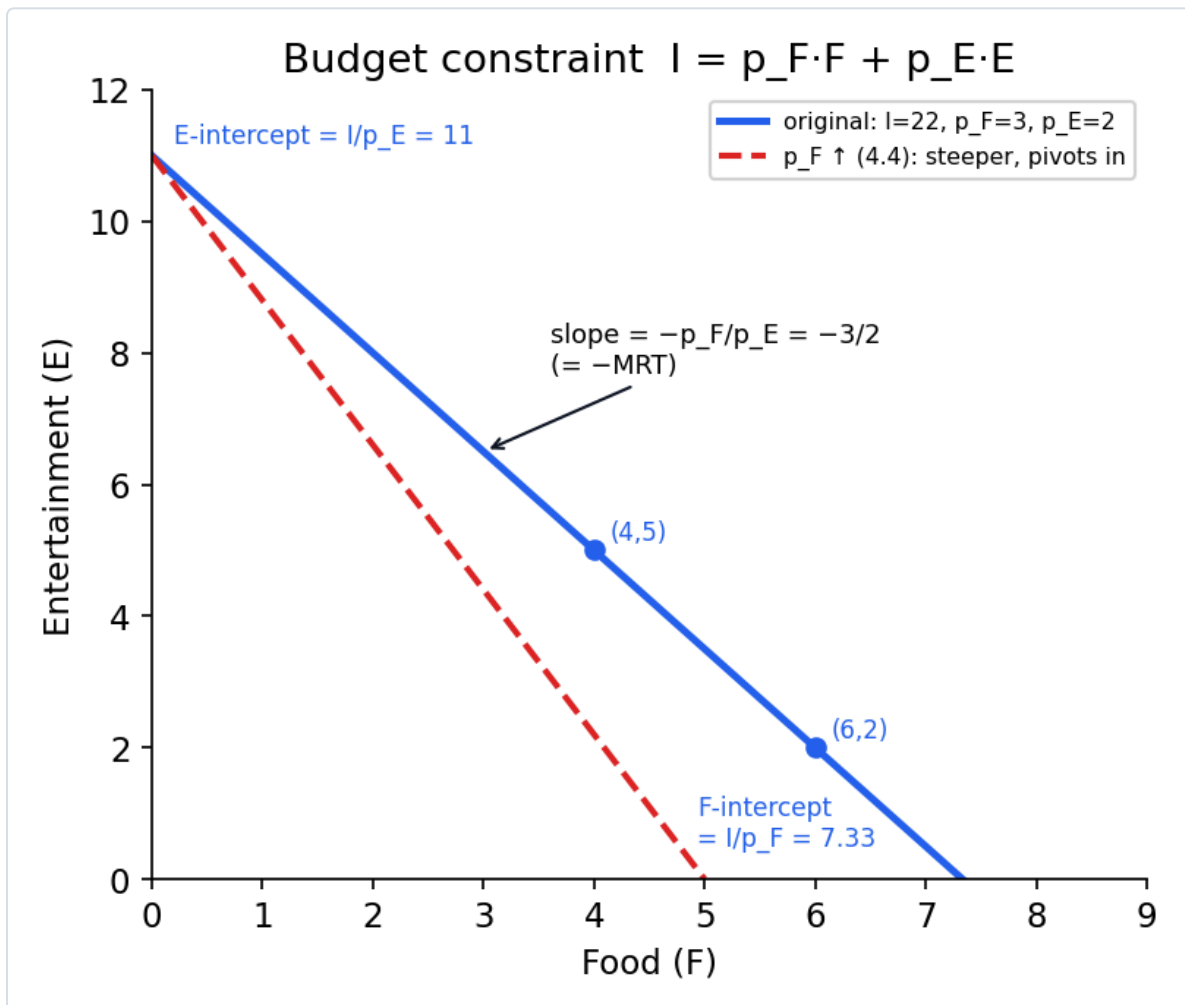
## §5.3 Budget constraint & the MRT (deck slides 21–22)

**Core:** the **budget constraint** operationalizes "you can't have it all": given prices and income  $I$ , total spending can't exceed income. With two goods it's a **straight line** (the **budget line**); everything on or below it is the **opportunity set**.

### BUDGET CONSTRAINT & LINE

$$I = p_F F + p_E E \implies E = I/p_E - (p_F/p_E) \cdot F$$

*Reads:* income = spending on Food + spending on Entertainment; rearranged for graphing, E-intercept =  $I/p_E$  (all income on E), slope =  $-p_F/p_E$ , F-intercept =  $I/p_F$  (all income on F).



- **Worked example (deck:  $I=22, p_F=3, p_E=2$ ):** E-intercept =  $22/2 = 11$ ; slope =  $-3/2$ ; F-intercept =  $22/3 \approx 7.33$ . Bundle  $(F=4, E=5)$  is affordable:  $3 \cdot 4 + 2 \cdot 5 = 22$  ✓. So is  $(F=6, E=2)$ :  $3 \cdot 6 + 2 \cdot 2 = 22$  ✓.

- **MRT — marginal rate of transformation** = the **slope of the budget line** = how the **market** lets you swap goods:

#### MARGINAL RATE OF TRANSFORMATION

$$MRT = p_F / p_E \quad (\text{the relative price = the market "exchange rate" between the goods})$$

*Reads:* give up one F, the market gives you  $p_F/p_E$  units of E. With  $p_F=3$ ,  $p_E=2$ ,  $MRT = 1.5$ : one less Food frees €3, which buys 1.5 Entertainment. (*MRS = what you're willing to trade; MRT = what the market lets you trade.*)

- **Comparative statics of the line (deck):**
  - **Income change** → **parallel shift** (slope unchanged; both intercepts move).
  - **Price change** → **pivot** (steepness changes). A higher  $p_F \Rightarrow$  **steeper**, pivoting **inward** around the E-intercept (see red dashed line); a lower  $p_F \Rightarrow$  **flatter**.

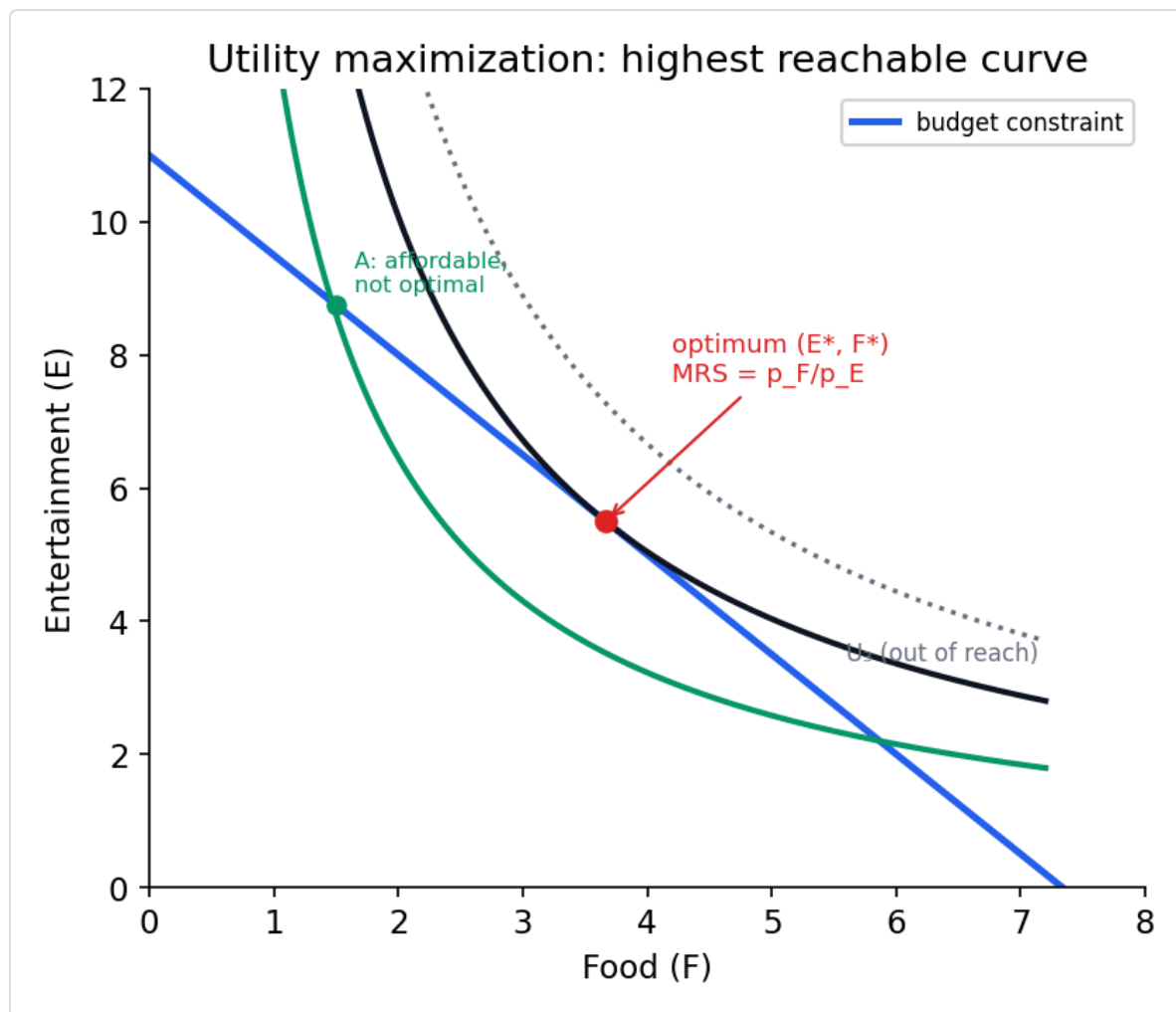
### ★ §5.4 The optimum: maximize utility subject to the budget (deck slide 23)

*Core:* the consumer solves **max  $U(F,E)$  subject to  $p_F \cdot F + p_E \cdot E \leq I$** . Graphically: find the **highest indifference curve** that still touches the budget line — that's the **tangency point**. At the tangency, the slope of the indifference curve (**MRS**) equals the slope of the budget line (**MRT**).

#### CONSUMER OPTIMUM (TANGENCY)

$$MRS = MRT \iff MU_F/MU_E = p_F/p_E \iff MU_F/p_F = MU_E/p_E$$

*Reads:* at the optimum, **the rate you're willing to trade = the rate the market lets you trade**. The last form is the "**equal marginal utility per euro**" (bang-for-the-buck) rule: **the last euro spent on Food yields the same extra utility as the last euro spent on Entertainment** — otherwise you'd reallocate toward the higher-MU-per-euro good.



**How to read it (utility maximization).** Point A is *affordable* (on the budget line) but **not optimal**: the curve through A is steeper than the line ( $MRS > MRT$ ), so she values Food more than the market charges → she should **buy more Food, less Entertainment**, sliding down to the tangency. The **red optimum** is the best reachable bundle; the dotted curve  $U_3$  would be better but her income can't reach it.

- **Why A isn't optimal (the deck's logic):** at A,  $MU_F/MU_E > p_F/p_E$  — food is "cheap relative to how much you value it." Reallocate toward food until the two ratios are equal.
- **Worked example ( $U = \sqrt{EF}$ ),  $I=22$ ,  $p_F=3$ ,  $p_E=2$ ):**  $MRS = E/F$ . Set  $MRS = MRT$ :  $E/F = 3/2$  →  $E = 1.5F$ . Plug into the budget  $3F + 2E = 22$ :  $3F + 2(1.5F) = 22$  →  $6F = 22$  →  $F^* = 3.67$ ,  $E^* = 5.5$ . (Check:  $3 \cdot 3.67 + 2 \cdot 5.5 = 11 + 11 = 22$  ✓ — matches the red dot in the figure.)

## Comparative statics · interior vs corner · the Cobb-Douglas shortcut (deck slide 24, §5.7)

**Core:** once you can find the optimum, you can ask **how it moves** when a price or income changes — and that movement, traced out, **is the demand curve** (Ch. 6). Watch for **corner solutions**.

- **Interior vs corner solution (deck):**
  - **Interior solution** — the tangency lies **strictly inside** the axes (buy *some* of both goods). Requires **convex** (imperfect-substitute) preferences.
  - **Corner solution** — the optimum is **on an axis** (buy *only one* good). Happens with **perfect substitutes**: if  $MRS \neq MRT$  everywhere, you spend **everything** on whichever good gives more

utility per euro ( $MU/p$ ). Example: if  $U = 2F + E$  and  $p_F = p_E$ , Food gives twice the bang per euro  $\rightarrow$  buy only Food.

- **Perfect complements** — always a corner-of-the-L: buy the goods in their **fixed ratio** regardless of the price ratio (solve the ratio + the budget line together).
- **The Cobb-Douglas shortcut (worth memorising for exams):** for  $U = F^a \cdot E^b$ , the tangency condition solves in closed form — the consumer **spends a constant share of income on each good**:

#### COBB-DOUGLAS OPTIMAL DEMANDS

$$F^* = \left[ \frac{a}{a+b} \right] \cdot I/p_F \quad \cdot \quad E^* = \left[ \frac{b}{a+b} \right] \cdot I/p_E$$

*Reads:* spend the fraction  $a/(a+b)$  of income on Food and  $b/(a+b)$  on Entertainment, then divide by each price to get quantities. The exponents are the expenditure shares. **Worked ( $U = F^{0.5}E^{0.5}$ )**, so  $a=b=0.5$ ,  $I=22$ ,  $p_F=3$ ,  $p_E=2$ :  $F^* = 0.5 \cdot 22/3 = 3.67$ ,  $E^* = 0.5 \cdot 22/2 = 5.5$  — same answer as the tangency method above, in one line. (Notice each good's demand **doesn't depend on the other good's price** — a special Cobb-Douglas feature.)

- **The appendix method (§5.7) — Lagrange:** for a general  $U$ , combine objective + constraint into  $\mathcal{L} = U(F,E) + \lambda(I - p_F F - p_E E)$ , set the three partials to zero, and the first two give back  $MU_F/MU_E = p_F/p_E$  — the same tangency condition.  $\lambda$  = the **marginal utility of income** ("how much extra utility one more euro of income buys"). *You won't need Lagrange for the exam, but know that it reproduces  $MRS = MRT$ .*
- **The payoff (Ch. 6 foreshadow):** change  $p_F$ , re-solve the optimum, and record the new  $F^*$  — repeat, and the  $(p_F, F^*)$  pairs **trace out the demand curve**. *This is why we built all this machinery: choice  $\rightarrow$  demand.*

## §5.5–5.6 Discussion of the framework & behavioral economics

**Core:** the rational-choice model is powerful (Becker: "maximizing behavior, market equilibrium, and stable preferences, used relentlessly and unflinchingly") — but it's a **model**. Two clarifications, then the **deviations**.

- **Two common misconceptions (§5.5):**
  - "Economics says be a selfish materialist." **No** — the framework maximizes **utility**, not *goods*. Giving to charity, choosing a low-paid academic job you love, buying organic — all perfectly rational if that's your preference.
  - "It only works for apples-vs-oranges." Actually **Gary Becker** extended it to fertility, marriage, crime, discrimination: assume **stable preferences** and let **costs/benefits differ** across contexts (e.g. why fertility differs across countries — Table 5.2 — need not be different *tastes*).
- **Behavioral economics (§5.6) — three deviations from the simple model:**

| Deviation                                 | What it is                                                                                                        | Signature examples                                                                                                                                                                                                                  |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Social preferences</b>                 | We care about <b>others</b> — fairness, status, altruism, spite. <i>Not irrational</i> — just richer preferences. | <b>Ultimatum game</b> : proposers offer ~40% (not ~0%), responders reject ~16% of low offers to punish unfairness.                                                                                                                  |
| <b>Lack of self-control</b>               | We over-weight the <b>present</b> — a myopic " <i>doer</i> " vs a " <i>planner</i> " (Thaler–Shefrin).            | Smoking, over-spending, under-saving; the marshmallow test; policy responses = mandatory pensions, alcohol limits.                                                                                                                  |
| <b>Cognitive limitations &amp; biases</b> | Systematic (not random) errors; <b>bounded rationality</b> , heuristics.                                          | <b>Loss aversion / status-quo bias / endowment effect</b> (Cornell coffee-mug study: owners value the mug 2–4× more); <b>framing</b> (600-lives problem); <b>overconfidence</b> (68% of entrepreneurs think they'll beat the odds). |

- **Random error vs bias (deck's dartboard)**: *random* misses scatter around the bull's-eye (people are right *on average*); a **behavioral bias** is a **systematic** skew off-centre (Kahneman & Tversky, *Thinking, Fast and Slow*).
- **Final word (§5.6.5)**: the rational model stays the **benchmark** — you need it *even to define* a "deviation." Behavioral economics tells you **when** the benchmark will mislead (e.g. demand for addictive painkillers vs baking soda).

## Cases & examples

- **Why did shrimp consumption triple? (Freakonomics / Levitt) — §5.4.1**. Non-economists guessed **demand shifts** (health fads, movies, trendy chefs). The choice framework points first to **prices and income**: the **relative price of shrimp fell** — that alone moves the optimum along/across the budget line. *Lesson: shifts in the budget constraint, not just tastes, drive choices.*
- **Same preferences, different choices: US pizza vs French duck confit (Dubois et al. 2014) — Fig 5.8**. Give an American and a French consumer the **identical** indifference map. Pizza is **cheaper in the US**, confit **cheaper in France** → different **budget-line slopes** → the American picks more pizza, the French more confit. *Different choices need not mean different tastes — just different relative prices.*
- **The ultimatum game — fairness is real (Güth 1982; Henrich 2001)**. Split €10; responder can veto (then both get €0). Pure self-interest predicts "offer €0.10, accept." Instead proposers offer **~40%**, and low offers get **rejected** — even across 15 small-scale societies worldwide (offers never below ~25%). *Evidence for social preferences.*
- **The Cornell coffee-mug experiment — the endowment effect (Kahneman et al. 1990)**. Mugs handed out at random; owners then valued them **2–4× higher** than non-owners would pay. Ownership shouldn't matter to a rational valuation — but a **reference point** does. *Loss aversion / status-quo bias in one image.*
- **Food stamps or cash? (deck slide 25)**. ~50% of US kids use food stamps before age 20. In the standard model, **cash is weakly better** (it relaxes the budget line *more* — you can buy food *or* anything else), so an in-kind transfer can only tie or lose. Yet food stamps persist — because of **paternalism / self-control** concerns and political constraints. *A perfect budget-constraint application with a behavioral twist.*

- **Fertility rates: tastes or costs? (Becker; Table 5.2).** South Korea 0.97 kids/woman vs Mali 6.14. A Becker-style reading: **same preferences**, different **costs/benefits** of children (childcare compatibility with careers, old-age support, child mortality, pensions). *Rational choice reaches far beyond "goods."*

## Formula sheet (Consumer Choice)

| Concept                                          | Formula                                                                             | In words                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Utility represents preferences</b>            | $a > b \Leftrightarrow U(a) > U(b)$                                                 | numbers rank bundles ( <b>ordinal</b> )                  |
| <b>Marginal utility</b>                          | $MU_F = \partial U / \partial F$                                                    | extra utility per extra unit; usually <b>diminishing</b> |
| <b>MRS (slope of indifference curve)</b>         | $MRS = MU_F / MU_E$                                                                 | how much E you'll trade for one more F, same utility     |
| <b>Budget line</b>                               | $I = p_F \cdot F + p_E \cdot E \rightarrow E = I/p_E - (p_F/p_E)F$                  | what you can afford; slope $-p_F/p_E$                    |
| <b>MRT (slope of budget line)</b>                | $MRT = p_F / p_E$                                                                   | how the <b>market</b> lets you trade F for E             |
| <b>Consumer optimum</b>                          | $MRS = MRT \Leftrightarrow MU_F/MU_E = p_F/p_E \Leftrightarrow MU_F/p_F = MU_E/p_E$ | tangency = equal MU per euro                             |
| <b>Cobb-Douglas demands</b><br>( $U = F^a E^b$ ) | $F^* = a/(a+b) \cdot I/p_F, E^* = b/(a+b) \cdot I/p_E$                              | spend constant income shares                             |

## Key terms

| Term                                    | One-line meaning                                                        |
|-----------------------------------------|-------------------------------------------------------------------------|
| <b>Bundle</b>                           | a specific quantity of each good                                        |
| <b>Completeness</b>                     | can rank <i>any</i> two bundles                                         |
| <b>Transitivity</b>                     | $A > B, B > C \Rightarrow A > C$ (rankings "hold in transit")           |
| <b>Monotonicity</b>                     | "more is better" — farther-out curves preferred                         |
| <b>Utility function</b>                 | numbers on bundles preserving the ranking ( <b>ordinal</b> )            |
| <b>Marginal utility (MU)</b>            | $\partial U / \partial(\text{good})$ ; extra utility from one more unit |
| <b>Indifference curve</b>               | all bundles of equal utility                                            |
| <b>MRS</b>                              | slope of indifference curve = $MU_F / MU_E$                             |
| <b>Budget constraint / line</b>         | $I = p_F F + p_E E$ ; the opportunity set                               |
| <b>MRT</b>                              | slope of budget line = $p_F / p_E$                                      |
| <b>Consumer optimum</b>                 | highest reachable curve tangent to the budget line                      |
| <b>Interior / corner solution</b>       | some of both goods / all of one good                                    |
| <b>Behavioral economics</b>             | systematic deviations from the rational model                           |
| <b>Loss aversion / endowment effect</b> | losses hurt more than equal gains; owning raises valuation              |

## Exam pointers

- **Draw the machine, then solve it.** For "find the optimal bundle": (1) write  $MRS = MU_F / MU_E$  , (2) set  $MRS = p_F / p_E$  , (3) combine with the **budget line**  $I = p_F F + p_E E$  , (4) solve the two equations. *Two equations, two unknowns — always.*
- **Cobb-Douglas → use the shortcut.** If  $U = F^a E^b$  , jump straight to  $F^* = a / (a+b) \cdot I / p_F$  . Saves time; derive it once so you trust it.
- **Check the second good.** After finding one quantity, plug back into the budget line for the other, and **verify it satisfies the constraint** ( $p_F F^* + p_E E^* = I$  ).
- **MRS = MU of the x-axis good over MU of the y-axis good.** Getting the ratio upside-down is the #1 error. If Food is on the x-axis,  $MRS = MU_F / MU_E$  and the tangency is  $MU_F / MU_E = p_F / p_E$  .
- **Corner solutions:** perfect substitutes → compare  $MU_F / p_F$  vs  $MU_E / p_E$  and spend everything on the winner. Perfect complements → ignore the price ratio, use the **fixed ratio + budget line**.
- **Comparative statics language:** income change = **parallel shift**; own-price change = **pivot**. Re-solving the optimum after a price change is how you **derive a demand curve** (Ch. 6).
- **Ordinal, not cardinal:** never say "twice as happy." Utility numbers only **rank**.
- **Behavioral:** be able to name the bias from a scenario (endowment effect, loss aversion, framing, self-control, overconfidence) and say it's a **systematic** (not random) deviation.

**Bridge to Lecture 5 (Ch. 6 — Demand curves & elasticities).** We now have a consumer who picks the tangency bundle for *given* prices and income. **Change one price and re-solve**, over and over, and the optimal quantities trace out a **demand curve** — closing the loop back to the demand we assumed in Lecture 2, and connecting to the **elasticities** of Lecture 3.

## ★ Practice problems (with solutions)

A full-chapter set: preferences & axioms → utility & MU → MRS → budget line & MRT → the optimum → Cobb-Douglas shortcut → corner solutions → comparative statics → behavioral. Try each before opening the solution. Use **Food (F)** on the x-axis, **Entertainment (E)** on the y-axis throughout.

1. A consumer has bundles  $A = (E\ 4, F\ 2)$ ,  $B = (E\ 8, F\ 2)$ ,  $C = (E\ 3, F\ 3)$ ,  $D = (E\ 6, F\ 5)$ . Using only "**more is better**," which rankings can you establish for certain, and which need the indifference map? (Compare A–B, C–D, and A–C.)
2. Preferences satisfy completeness, transitivity, monotonicity. A student says: "*I prefer coffee to tea, tea to cocoa, but cocoa to coffee.*" Which **axiom** does this violate, and why is it a problem for building a utility function?
3.  $U = \sqrt{EF}$ . (a) Find the utility of  $(E\ 2, F\ 8)$ ,  $(E\ 4, F\ 4)$ ,  $(E\ 8, F\ 2)$ ,  $(E\ 5, F\ 5)$ . (b) Which lie on the same indifference curve? (c) Rank all four bundles.
4.  $U = F^{0.5} E^{0.5}$ . (a) Derive  $MU_F$  and  $MU_E$ . (b) Evaluate both at  $(F\ 4, E\ 9)$ . (c) Does  $MU_F$  rise or fall as F increases (E fixed)? Name the principle.
5. For  $U = \sqrt{EF}$ , show that  $MRS = E/F$ . Then compute the MRS at  $(E\ 2, F\ 8)$  and at  $(E\ 8, F\ 2)$  and interpret the difference in one sentence.
6. Income  $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ . (a) Write the budget line as  $E = \dots$ . (b) Give both intercepts and the slope. (c) Is  $(F\ 6, E\ 8)$  affordable? (d) What is the **MRT**, and what does it mean?
7. Same as 6 ( $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ ). (a) The government gives a **€20 income top-up**. What happens to the budget line? (b) Instead,  $p_F$  falls to 2. What happens? (c) Sketch the difference in words (shift vs pivot).
8.  $U = \sqrt{EF}$ ,  $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ . Find the **optimal bundle** using  $MRS = MRT$  + the budget line. Verify it satisfies the constraint.
9.  $U = F^{0.25} E^{0.75}$  (Cobb-Douglas),  $I = 60$ ,  $p_F = 3$ ,  $p_E = 6$ . Use the **expenditure-share shortcut** to find  $F^*$  and  $E^*$ . What share of income goes to each good?
10. (Perfect substitutes.)  $U = 2F + E$ ,  $I = 24$ ,  $p_F = 3$ ,  $p_E = 2$ . (a) Compute  $MU_F/p_F$  and  $MU_E/p_E$ . (b) What is the optimal bundle, and why is it a **corner solution**?
11. (Perfect complements.)  $U = \min(F, E)$ ,  $I = 30$ ,  $p_F = 2$ ,  $p_E = 3$ . Find the optimal bundle. (Hint: the optimum sits at the corner where  $F = E$ ; combine with the budget line.)
12. (Comparative statics.) Start from problem 8's optimum ( $U = \sqrt{EF}$ ,  $I=40$ ,  $p_F=4$ ,  $p_E=2$ ). Now  $p_F$  falls to €2. (a) Find the new optimum. (b) Did the quantity of Food rise or fall? (c) You now have **two  $(p_F, F^*)$  points** — what curve are you tracing out?
13. (Behavioral.) Name the bias in each: (a) You keep a concert ticket you'd never pay today's resale price for. (b) "1 in 3 patients survive" feels better than "2 in 3 die." (c) You put your alarm clock across the

room so you can't hit snooze. (d) 80% of new founders rate their success odds  $\geq 7/10$ .

## Solutions

1. **A vs B:** B has more E (8 vs 4), same F  $\rightarrow B > A$  (more is better). **C vs D:** D has more of *both* (E 6 > 3, F 5 > 3)  $\rightarrow D > C$ . **A vs C:** A has more E (4 > 3) but less F (2 < 3)  $\rightarrow$  **cannot rank** by "more is better" alone; you need the **indifference map** (depends on the MRS).
2. Violates **transitivity**. coffee > tea and tea > cocoa should imply coffee > cocoa, but she says cocoa > coffee — the preferences **cycle**. A cycling ranking **can't be represented by a utility function** (no set of numbers can have coffee > tea > cocoa > coffee), so the whole model of choice breaks down.
3. (a)  $U = \sqrt{EF}$ : (E2,F8)  $\rightarrow \sqrt{16} = 4$ ; (E4,F4)  $\rightarrow \sqrt{16} = 4$ ; (E8,F2)  $\rightarrow \sqrt{16} = 4$ ; (E5,F5)  $\rightarrow \sqrt{25} = 5$ . (b) The **first three** (all  $E \cdot F = 16$ ) lie on the **same indifference curve** ( $U=4$ ). (c) **(E5,F5) > {the other three, which are indifferent to each other}** — i.e.  $(5,5) > (2,8) \sim (4,4) \sim (8,2)$ .
4. (a)  $MU_F = \partial U / \partial F = 0.5 F^{-0.5} E^{0.5} = 0.5 \sqrt{E/F}$ ;  $MU_E = 0.5 F^{0.5} E^{-0.5} = 0.5 \sqrt{F/E}$ . (b) At (F4,E9):  $MU_F = 0.5 \sqrt{9/4} = 0.5 \cdot 1.5 = 0.75$ ;  $MU_E = 0.5 \sqrt{4/9} = 0.5 \cdot 0.667 = 0.333$ . (c) As F rises,  $\sqrt{E/F}$  falls  $\rightarrow MU_F$  falls = **diminishing marginal utility**.
5.  $MRS = MU_F / MU_E = [0.5 \sqrt{E/F}] / [0.5 \sqrt{F/E}] = \sqrt{E/F} \cdot \sqrt{E/F} = E/F$ . At (E2,F8):  $MRS = 2/8 = 0.25$ ; at (E8,F2):  $MRS = 8/2 = 4$ . **Interpretation:** with lots of F she'll give up only 0.25 E for one more F; with little F she'll give up 4 E — the MRS **falls** as F rises (diminishing MRS / convex curve).
6. (a)  $40 = 4F + 2E \rightarrow E = 20 - 2F$ . (b) E-intercept = 20, F-intercept = 10, slope = -2. (c) (F6,E8):  $4 \cdot 6 + 2 \cdot 8 = 24 + 16 = 40 \rightarrow$  **exactly affordable** (on the line). (d)  $MRT = p_F / p_E = 4/2 = 2 \rightarrow$  giving up 1 Food frees €4, which buys 2 Entertainment.
7. (a) **Income +€20** ( $I = 60$ ): **parallel outward shift** — new intercepts  $E = 60/2 = 30$ ,  $F = 60/4 = 15$ ; **slope unchanged** (-2). (b)  **$p_F$  falls to 2:** budget **pivots** — E-intercept **unchanged at 20**, F-intercept rises to  $40/2 = 20$ , slope flattens to  $-p_F / p_E = -1$ . (c) Income change = **parallel shift** (same slope, more of everything); own-price change = **pivot** (one intercept fixed, slope changes).
8.  $MRS = E/F$  (from problem 5). Set  $MRS = MRT$ :  $E/F = 4/2 = 2 \rightarrow E = 2F$ . Budget:  $4F + 2E = 40 \rightarrow 4F + 2(2F) = 40 \rightarrow 8F = 40 \rightarrow F^* = 5, E^* = 10$ . (Check:  $4 \cdot 5 + 2 \cdot 10 = 20 + 20 = 40$  ✓.)
9. Shortcut for  $U = F^{0.25} E^{0.75}$  ( $a=0.25, b=0.75, a+b=1$ ):  $F^* = 0.25 \cdot 60/3 = 0.25 \cdot 20 = 5$ ;  $E^* = 0.75 \cdot 60/6 = 0.75 \cdot 10 = 7.5$ . **Shares:** 25% of income (€15) on Food, 75% (€45) on Entertainment. (Check:  $3 \cdot 5 + 6 \cdot 7.5 = 15 + 45 = 60$  ✓.)
10.  $U = 2F + E$ :  $MU_F = 2$ ,  $MU_E = 1$ . (a)  $MU_F / p_F = 2/3 \approx 0.67$ ;  $MU_E / p_E = 1/2 = 0.5$ . (b) Food gives **more utility per euro** ( $0.67 > 0.5$ ) and the rate is **constant** (perfect substitutes), so spend **everything on Food**:  $F^* = I / p_F = 24/3 = 8$ ,  $E^* = 0$ . **Corner solution** because  $MU_F / p_F \neq MU_E / p_E$  everywhere — there's no interior tangency; the highest indifference (straight) line touches the budget line **on the F-axis**.
11.  $U = \min(F, E)$ : no point in having more of one than the other, so the optimum has  **$F = E$** . Budget:  $2F + 3E = 30$  with  $E = F \rightarrow 2F + 3F = 30 \rightarrow 5F = 30 \rightarrow F^* = E^* = 6$ . (Check:  $2 \cdot 6 + 3 \cdot 6 = 12 + 18 = 30$  ✓.) (Note: the price ratio is irrelevant to the ratio of goods — only to how much of the fixed bundle you can afford.)

12. Now  $p_F = 2$ ,  $p_E = 2$ ,  $I = 40$ .  $MRS = E/F = p_F/p_E = 2/2 = 1 \rightarrow E = F$ . Budget:  $2F + 2E = 40$  with  $E = F \rightarrow 4F = 40 \rightarrow F^* = 10, E^* = 10$ . (b) Food rose from 5 to 10 as its price fell  $4 \rightarrow 2$  (a downward-sloping demand response). (c) The two points ( $p_F=4, F=5$ ) and ( $p_F=2, F=10$ ) are two points on the consumer's **demand curve for Food**.

13. (a) **Endowment effect / sunk-cost + status-quo bias** (you value it more because you own it). (b) **Framing effect** (same facts, "survive" vs "die"). (c) **Lack of self-control** — a commitment device (planner constrains the future doer). (d) **Overconfidence**.

## ★ Fun facts & memorable details (Lecture 4)

Sticky bits from Consumer Choice.

- **"More is better" has a loophole:** economists just **redefine bads as the absence of the bad** — you don't demand pollution, you demand *clean air*. A neat trick that keeps monotonicity intact.
- **Utility is unobservable — so numbers lie.**  $U = 8$  is not "twice as good" as  $U = 4$ ; utility is purely **ordinal**. Early economists agonised over this; the modern answer is "we only ever need the ranking."
- **The bang-for-your-buck rule is the whole optimum.**  $MU_F/p_F = MU_E/p_E$  — at the best bundle, the **last euro** spent on each good buys the **same** extra utility. If it didn't, you'd move money to the better-value good. That's *all* tangency means.
- **Cobb-Douglas people are creatures of habit:** with  $U = F^a E^b$  you always spend the **same share** of income on each good — no matter how rich you get or how prices move. That's why it's the field's favourite functional form.
- **Left shoes are worthless alone.** Perfect complements (L-shaped curves) are why a single earring, one glove, or a lone right shoe adds **zero** utility — you need the matching unit.
- **The ultimatum game breaks "Homo economicus."** Offered a lopsided split, people **pay real money to punish unfairness** — and this holds from Pittsburgh to Papua New Guinea. Fairness is not a Western quirk.
- **You'd sell your coffee mug for far more than you'd pay for it.** The **endowment effect**: Cornell students valued a randomly-given mug **2–4× higher** than students who didn't get one. Ownership rewires valuation.
- **The paradox of voting:** a strict cost-benefit reading says *don't vote* (your vote almost never decides the election). That people vote anyway is a clue that **narrow self-interest is too small a theory of us**.
- **Ulysses tied himself to the mast** to resist the Sirens — a 3,000-year-old **commitment device**, the same logic as not keeping cigarettes in the house. Self-control problems are ancient.